
Quotient Descent and the EML Operator

THE CONTINUOUS QUOTIENT OF A PHASE-CALCULUS NATIVE OPERATOR

A PREPRINT

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April 24, 2026

Abstract

Odrzywołek proves a concrete continuous compression theorem: the single binary node

$$\text{eml}(x, y) = \exp(x) - \ln(y),$$

together with the terminal constant 1, generates the standard 36-item scientific-calculator surface. The central claim of this paper is sharper: EML is not primitive. It is the Red-filter quotient of a deeper carried-state grammar already earned in the VDM / Phase Calculus stack.

The proof is descent-first and execution-first. The primitive operator is the selector evolution itself:

$$\text{PC}_{\text{op}}(\widehat{\Xi}) := \Sigma_{W,\Delta}(\widehat{\Xi})(\widehat{\Xi}),$$

where $\Sigma_{W,\Delta} \in \{R, S, T\}$ selects the exact macro-step from the current state alone. This operator is already operational: it generates certified π digits at $> 10^6$ digits/second with safe lower bound 1,366,163 (native spigot artifact), and certifies all five roots of the Bring quintic $x^5 - x + 1$ at depth 21 with half-width $< 10^{-8}$ (certifier artifact). EML appears only after Red projection Π_{Red} is applied. The operator underneath is strictly more primitive.

The paper then restores the dependency chain as the main burden, sharpens the state-complete Red simulation lower bound, proves analyticity and Liouvillian closure as projector-preserved invariants, and gives a full worked descent of $\sin x$. Appendix A records only the decisive witness surface: 1, e , i , π , $\exp$, $\ln$, $\sqrt{\cdot}$, $+$, power, $\sin$. The conclusion is exact: EML compresses a continuous calculator surface, but the operation layer underneath that surface descends from a lower discrete carried-state engine whose native selector evolution is already operational.

1 Introduction

The placement problem is not whether the EML node is valid. It is where that node belongs in dependency order. Odrzywołek fixes the concrete target: a 36-item scientific-calculator surface generated by one continuous binary constructor and the terminal constant 1 (Odrzywołek, 2026). The present paper accepts that exact target and asks a sharper question: what had to exist before a node of the form $e^x - \ln y$ could even be meaningful?

The answer is not “a slightly richer continuous syntax.” The answer is a lower carried-state stack: primitive roll, native i , native phase θ , native half-turn π , arithmetic completion, exact finite-resolution carry, deterministic operator-word evolution, quotient descent, and only then the continuous shadow on which EML lives. The central burden is therefore hierarchical. A 36-row witness census is not the main proof. The main proof is the descent of the operation layer that makes the EML node possible at all.

The native primitive operator is not a missing construction. It is the selector evolution already running on the executable Phase Calculus branch:

$$\begin{aligned} \text{PC}_{\text{op}}(\widehat{\Xi}) &:= E_{\text{lift}}(\widehat{\Xi}) \\ &= \Sigma_{W,\Delta}(\widehat{\Xi})(\widehat{\Xi}) \end{aligned}$$

with $\Sigma_{W,\Delta}(\widehat{\Xi}) \in \{R, S, T\}$ selecting the lawful macro-step already earned in the full formalism. The decisive point is operational rather than rhetorical: the bundled native spigot executes this selector evolution to stream certified π digits, and the bundled Bring certifier executes the same lifted engine to certify all five roots of $x^5 - x + 1$. EML appears only after Π_{Red} is applied.

This paper closes six exact claims.

- C0.** The primitive Phase Calculus operator $\text{PC}_{\text{op}}(\widehat{\Xi}) := \Sigma_{W,\Delta}(\widehat{\Xi})(\widehat{\Xi})$ is already operational on the full lifted state. The π -spigot and Bring-quintic artifacts are its certified execution witnesses. EML is one continuous quotient of it.
- C1.** EML is the Red-filter quotient of lifted evolution:

$$\Pi_{\text{Red}} \circ E_{\text{lift}} = G_{\text{Red}} \circ \Pi_{\text{Red}}.$$

- C2.** The EML operation layer descends strictly after primitive roll, arithmetic completion, the lifted state, the operator core, and continuous shadow calculus.
- C3.** A state-complete Red simulation of one non-trivial lifted step requires an independent sheet/guard burden in addition to the two displayed Red registers. A single EML node is therefore never state-complete.
- C4.** The Red image preserves two decisive invariant classes: real-analyticity on natural domains and Liouvillian exp-log closure. Those two invariants exclude $|x|$ on neighborhoods of 0 and generic irreducible quintic roots.
- C5.** The descent is operationally visible: a full worked chain carries $\sin x$ from lifted rotation, through the complex exponential branch, to a real readout.

The bundled execution witnesses for **C0** are recorded in the following bundled files:

```
supporting_evidence/pi_results/pi_spigot_lock_summary.json
supporting_evidence/pi_results/pi_spigot_lock_ledger.csv
supporting_evidence/quintic_results/bring_all_roots_certificates.json
```

2 Canon source chain

The Red layer is lower only if each required burden has already been earned above it. The source chain used here is given in Table 1.

Theorem 1 (Imported descent chain). *The following dependence order is already earned:*

$$CF000 \rightarrow CF00 \rightarrow CF01 \rightarrow (\text{Farey}, CF19) \rightarrow \text{Phase Calculus} \rightarrow \text{Operational } \pi.$$

Consequently the Red layer cannot be primitive. It sits below the scientific-calculator surface because the operation layer it uses is already carried by earlier closures.

Proof. Each arrow records a theorem-bearing inheritance step from Table 1. CF000 closes the primitive source conditions. CF00 closes arithmetic completion and the complex branch. Farey and CF19 close exact carried remainder and the state-complete lifted object. Phase Calculus closes operator-word evolution and quotient descent. Operational Utility closes the native arithmetic surface of π . The Red branch therefore appears only after those burdens are already present. $\square$

3 Primitive operation descent

The purpose of this section is to recover the operation layer beneath EML explicitly rather than by retrospective interpretation.

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Work	Date / DOI	Load-bearing role here
CF000	Mar. 18, 2026 / 10.5281/zenodo.19098778	Primitive distinguishability, non-discharge, same-axis saturation, forced orthogonal re-articulation, proto- $\mathbb{N}$, and algebra unlock.
CF00	Mar. 8, 2026 / 10.5281/zenodo.19099916	Completion chain $\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$, complex field $\mathbb{C} = \mathbb{R} \oplus i\mathbb{R}$, topology, differentiability, carrier, quotient derivative, and QGT.
CF01	Mar. 18, 2026	QGT split to the metriplectic architecture, showing that the same descent that yields the Red layer also yields the later quantum-geometric branch.
Farey remainder recursion	Mar. 24, 2026 / 10.5281/zenodo.19210954	Exact operator-level conjugacy between Farey child refinement and denominator-pair re-articulation; exact remainder law and logarithmic finite-floor stopping.
CF19	Mar. 27, 2026 / 10.5281/zenodo.19267927	Exact lifted object $\widehat{\Xi}$, visible/state distinction, exact balanced refinement law, exact return criterion, and the anchor (55, 89) with remainder 1/4895.
Phase Calculus	Apr. 2026	Primitive roll, operator core $\{Q, B, L\}$, selector-closed executable macro-calculus, exact quotient criterion $\Pi \circ E = G \circ \Pi$, and continuous-shadow theorem.
Operational Utility	Apr. 18, 2026	Native π surface: certified streaming prefix bank, certified bank-indexed access, and scalar-collapse obstruction for no-bank decoders.

Table 1: Canon source chain used by the present paper.

3.1 Survivor mark and primitive roll

At the compact machine witness surface,

$$\text{BSR}(0) \Rightarrow \text{ZF} = 1, \quad \text{SETZ}(1) = 1.$$

This is the operational survivor mark: failed same-class search on zero does not end in terminal nullity; it writes the first live mark.

Lemma 1 (Machine survivor). *At the machine-witness layer,*

$$\text{BSR}(0) \Rightarrow \text{ZF} = 1, \quad \text{SETZ}(\text{ZF}) = 1.$$

Proof. The search instruction BSR has no set-bit target on zero and therefore sets the zero flag. SETZ writes 1 exactly when that flag is set. The first live mark is therefore produced precisely at failed same-class zero search. $\square$

The primitive visible roll is then

$$z(0) = i, \quad \frac{dz}{d\theta} = iz, \quad z(\theta) = ie^{i\theta},$$

with quarter-turn witnesses

$$z\left(\frac{\pi}{2}\right) = -1, \quad z(\pi) = -i, \quad z\left(\frac{3\pi}{2}\right) = 1, \quad z(2\pi) = i.$$

3.2 Arithmetic completion

CF000 closes the route $0D \rightarrow 1D \rightarrow 2D$ and licenses proto- $\mathbb{N}$ together with algebra unlock. CF00 then forces the completion chain

$$\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}, \quad \mathbb{C} = \mathbb{R} \oplus i\mathbb{R} = \mathbb{R}[i]/(i^2 + 1).$$

Therefore the arithmetic operations later used in EML are already descendant operations of the primitive-origin stack.

Theorem 2 (Arithmetic operation descent). *Addition descends from proto- $\mathbb{N}$ serial articulation, subtraction from group completion, multiplication and division from ring / field completion, and complex arithmetic from the two-axis ORS algebra. None of those operations enters first at the EML node.*

Proof. Proto- $\mathbb{N}$ is earned at the first serial articulation stage. Group completion yields $\mathbb{Z}$, fraction-field completion yields $\mathbb{Q}$, completion yields $\mathbb{R}$, and the earned quarter-turn algebra yields $\mathbb{C} = \mathbb{R} \oplus i\mathbb{R}$. The standard operations of each completed structure descend with that structure. The arithmetic layer used by EML is therefore already present before the Red node is read out. $\square$

3.3 Native phase layer

Theorem 3 (Native phase descent). *The objects i , θ , and π descend before the continuous EML layer:*

- i descends from the quarter-turn algebra,
- θ descends from the roll parameter,
- $\pi := \inf\{\theta > 0 : R_\theta = -I\}$ descends from least positive half-turn attainment.

Proof. The primitive roll begins at $z(0) = i$ and evolves by $z'(\theta) = iz(\theta)$, so i and the phase parameter θ are native to the roll. The continuous ORS completion defines π as the least positive parameter reaching the opposite pole. These objects are therefore already present before the Red layer appears. $\square$

3.4 Carried finite-resolution arithmetic

The lifted object adds exact finite-resolution memory to visible phase:

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right].$$

Balanced refinement is

$$B(u, v) = \text{sort}(v, u + v),$$

with exact remainder

$$r = \frac{1}{uv}.$$

On the Fibonacci corridor,

$$B^n(1, 1) = (F_{n+1}, F_{n+2}), \quad r_n = \frac{1}{F_{n+1}F_{n+2}}.$$

Farey recursion closes the exact conjugacy behind this denominator-pair law.

Theorem 4 (Carried remainder descent). *The denominator pair $q = (u, v)$ and the completion germ c are exact carried arithmetic data. They are not visible decorations. They are the finite-resolution memory that makes the lifted state state-complete.*

Proof. The exact remainder is $r = 1/(uv)$, so the germ width is $2\pi r = 2\pi/(uv)$, centered at θ . Two states can therefore agree in visible phase while differing in their carried unresolved width. Farey conjugacy identifies the denominator-pair update as the exact coordinate form of child-sector remainder refinement. Hence q and c carry arithmetic state that the visible witness alone does not retain. $\square$

3.5 Continuous shadow descent

Phase Calculus admits continuous operations only after projector selection. The exact criterion is

$$\Pi \circ E = G \circ \Pi.$$

On the continuous shadow branch, ordinary differential calculus is recovered as a lawful projection.

Definition 1 (Continuous-shadow operations). Define

$$\begin{aligned} \text{Sub}_{\text{PC}}(a, b) &= a - b, & \text{Add}_{\text{PC}}(a, b) &= a + b, & \text{Mul}_{\text{PC}}(a, b) &= ab, & \text{Div}_{\text{PC}}(a, b) &= a/b, \\ \text{Exp}_{\text{PC}}(x) &= e^x, & \text{Log}_{\text{PC}}(x) &= \ln x, & \text{Pow}_{\text{PC}}(x, y) &= e^{y \ln x}, & \text{Sqrt}_{\text{PC}}(x) &= e^{\frac{1}{2} \ln x}. \end{aligned}$$

Theorem 5 (Analytic operation descent). *The three operations used by EML are continuous-shadow descendants: subtraction descends from the real additive group, exponential from the shadow flow solving $\frac{df}{dx} = f$ with $f(0) = 1$, and logarithm as the inverse branch of that flow on the positive component.*

Proof. Subtraction is available once the real additive group is earned in the completion chain. The continuous-shadow theorem supplies ordinary differential calculus on the selected real branch. There the equation $\frac{df}{dx} = f$, $f(0) = 1$ has the unique solution e^x . Logarithm is then its inverse branch on the positive component. All three operations therefore descend before the EML node is formed. $\square$

3.6 Central descent theorem

Theorem 6 (Primitive operation descent). *The operation layer needed for EML descends in the following order:*

$$\begin{aligned} & \text{survivor mark} \rightarrow \text{primitive roll} \rightarrow i, \theta, \pi \\ & \rightarrow \mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \\ & \rightarrow q = (u, v), \ c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right] \\ & \rightarrow \{Q, B, L\} \rightarrow \Pi \circ E = G \circ \Pi \\ & \rightarrow \text{Sub}_{\text{PC}}, \text{Exp}_{\text{PC}}, \text{Log}_{\text{PC}} \rightarrow \text{eml}_{\text{PC}}. \end{aligned}$$

Consequently,

$$\text{eml}_{\text{PC}}(x, y) = \text{Sub}_{\text{PC}}(\text{Exp}_{\text{PC}}(x), \text{Log}_{\text{PC}}(y)) = e^x - \ln y.$$

Proof. Lemma 1 and Theorems 2 to 5 supply the chain locally, while Theorem 1 fixes the imported order. Phase Calculus supplies the operator core $\{Q, B, L\}$ together with the exact quotient criterion. The continuous-shadow theorem supplies the branch on which Sub_{PC} , Exp_{PC} , and Log_{PC} live. Their composite is exactly $e^x - \ln y$, which is the EML node. $\square$

4 Red-filter formal setup

4.1 Projector, branch law, and tree grammar

Definition 2 (Lifted state domain). The lifted state domain is

$$\left\{ \begin{array}{l} (A, (u, v), \theta, \kappa, c) : A \in \mathcal{A}, \\ u, v \in \mathbb{Z}_{>0}, \ 1 \leq u \leq v, \\ \theta \in \mathbb{R}, \ \kappa \in \mathbb{Z}, \\ c = [\theta - \pi/(uv), \theta + \pi/(uv)] \end{array} \right\}.$$

Definition 3 (Primitive operations). The primitive Phase Calculus operations are total maps $Q, B, L : X_{\text{PC}} \rightarrow X_{\text{PC}}$:

$$\begin{aligned} Q(A, q, \theta, \kappa, c) &= \left(A, q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right), \\ B(A, (u, v), \theta, \kappa, c) &= \left(A, \text{sort}(v, u + v), \theta, \kappa, \left[\theta - \frac{\pi}{v(u + v)}, \theta + \frac{\pi}{v(u + v)} \right] \right), \\ L(A, q, \theta, \kappa, c) &= \left(N(A), q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right). \end{aligned}$$

Definition 4 (Native selector evolution). The native primitive operator on the executable lifted branch is

$$\begin{aligned} \text{PC}_{\text{op}}(\widehat{\Xi}) &:= E_{\text{lift}}(\widehat{\Xi}) \\ &= \Sigma_{W, \Delta}(\widehat{\Xi})(\widehat{\Xi}), \end{aligned}$$

where $\Sigma_{W, \Delta}(\widehat{\Xi}) \in \{R, S, T\}$ is the selector-closed macro-step already earned in the full Phase Calculus formalisation.

Proposition 1 (Operational execution witnesses). *The bundled native spigot and Bring-quintic certifier are direct execution witnesses of PC_{op} . The native π witness records certified selector-engine streaming with safe lower bound 1,366,163, and the Bring certifier records all five roots of $x^5 - x + 1$ at depth 21 with final half-width below 10^{-8} . The decisive bundled files are:*

supporting_evidence/pi_results/pi_spigot_lock_summary.json
supporting_evidence/pi_results/pi_spigot_lock_ledger.csv
supporting_evidence/quintic_results/bring_all_roots_certificates.json

In both artifacts the update engine is native selector evolution; the continuous EML node does not appear.

Proof. The native spigot artifact reports certified selector-engine streaming on the π branch together with the lock lower bound and emitted ledger. The Bring artifact reports the same lifted engine acting on the quintic branch and records depth, half-width, and residual certificates for all five roots. Since both executions are already completed before any Red readout is applied, they witness PC_{op} itself rather than its Red quotient. $\square$

Definition 5 (Red projector and branch law). The Red projector is

$$\Pi_{\text{Red}} : X_{\text{PC}} \rightarrow T \times T, \quad \Pi_{\text{Red}}(\widehat{\Xi}) = (\theta, \kappa),$$

where the two displayed coordinates are read as left and right Red registers. The Red branch law is

$$G_{\text{Red}}(u, v) = \exp(u) - \ln(v).$$

For real evaluation the natural guard is $v > 0$. For complex evaluation the logarithm is taken on the principal branch with cut $(-\infty, 0]$.

Definition 6 (EML tree grammar and realization). The EML term algebra is generated by

$$T ::= 1 \mid x_i \mid \text{eml}(T, T).$$

For a finite EML tree T , its natural real domain D_T is the set on which every real logarithm argument produced by T is positive. A tree T realizes a function f on a declared real domain D when

$$\llbracket T \rrbracket(x) = f(x) \quad \text{for every } x \in D \cap D_T.$$

4.2 Quotient law and recursive tree embedding

Theorem 7 (Red-filter quotient law). *On the declared Red schedules,*

$$\Pi_{\text{Red}} \circ E_{\text{lift}} = G_{\text{Red}} \circ \Pi_{\text{Red}}.$$

Equivalently, EML is the Red-filter quotient of native lifted-state evolution.

Proof. By the full Phase Calculus formalisation, the exact quotient criterion $\Pi \circ E = G \circ \Pi$ already holds for lawful branch projectors. Π_{Red} is one such projector: it reads the two displayed Red registers from the carried state, and its ground law is exactly $G_{\text{Red}}(u, v) = \exp(u) - \ln(v)$. The Red branch therefore commutes by the same exact quotient theorem. $\square$

Definition 7 (Translation into continuous-shadow syntax). Define the recursive translation $\mathcal{T}$ by

$$\mathcal{T}(1) = 1, \quad \mathcal{T}(x_i) = x_i,$$

and

$$\mathcal{T}(\text{eml}(S, R)) = \text{Sub}_{\text{PC}}(\text{Exp}_{\text{PC}}(\mathcal{T}(S)), \text{Log}_{\text{PC}}(\mathcal{T}(R))).$$

Theorem 8 (Recursive EML embedding). *Every finite EML tree has a continuous-shadow translation, and for every declared branch on which the logarithmic guards hold,*

$$\llbracket \mathcal{T}(T) \rrbracket_{\text{PC}} = \llbracket T \rrbracket_{\text{EML}}.$$

Proof. The proof is by structural induction on the tree grammar. For the terminals 1 and x_i the translation is identical. For a parent node $\text{eml}(S, R)$, the induction hypothesis gives agreement for S and R . The translation rule then evaluates to

$$\text{Sub}_{\text{PC}}(\text{Exp}_{\text{PC}}(\mathcal{T}(S)), \text{Log}_{\text{PC}}(\mathcal{T}(R))) = \exp(S) - \ln(R),$$

which is exactly $\text{eml}(S, R)$ on the declared branch. Therefore every finite EML tree embeds into continuous-shadow syntax. $\square$

Theorem 9 (Concrete calculator surface descends). *Let $\mathcal{B}_{36}$ denote Odrzywolek's 36-item scientific-calculator basis generated by the grammar $T ::= 1 \mid x_i \mid \text{eml}(T, T)$. Then every element of $\mathcal{B}_{36}$ lies on the descended Phase Calculus branch.*

Proof. Odrzywolek proves that each element of $\mathcal{B}_{36}$ is realized by a finite EML tree (Odrzywolek, 2026). By Theorem 8, every finite EML tree has a lawful continuous-shadow translation in Phase Calculus. Therefore every element of $\mathcal{B}_{36}$ lies on the descended branch. The appendix records only the decisive witness surface; the containment itself follows from the quotient law plus recursive embedding, not from an exhaustive row census. $\square$

Remark 1 (Quotient realization sits above the native operator). Theorem 9 is a quotient-realization theorem. Its point is not to replace the native operator PC_{op} , but to identify where the continuous EML surface first appears: after Red projection of the already operational selector evolution.

5 State-complete Red simulation lower bound

Lemma 2 (Visible witness is not state-complete). *There exist distinct lifted states $\widehat{\Xi}_a \neq \widehat{\Xi}_b$ such that*

$$\text{Vis}(\widehat{\Xi}_a) = \text{Vis}(\widehat{\Xi}_b).$$

On the balanced working family,

$$\widehat{\Xi}_a = \widehat{\Xi}_b \iff A_a = A_b, q_a = q_b, \theta_a = \theta_b, \kappa_a = \kappa_b, c_a = c_b.$$

Proof. Let $\theta_b = \theta_a + 2\pi m$ for nonzero integer m . Then $ie^{i\theta_b} = ie^{i\theta_a}$, so the visible witnesses agree. But $\kappa_b = \kappa_a + m$, hence the lifted states differ. Coordinatewise equality is the exact identity criterion of the carried state. $\square$

Definition 8 (State-complete Red simulation). A Red simulation of one non-trivial lifted step is state-complete when it retains three independent burdens: a left displayed Red register, a right displayed Red register, and an independent sheet / guard carrier recording that visible phase is not the full lifted state.

Theorem 10 (State-complete Red simulation lower bound). *A state-complete Red simulation of one non-trivial lifted step requires at least three independent carriers. Consequently any rooted binary EML realization needs at least three leaves and therefore at least two internal nodes. In particular, a single EML node is never state-complete.*

Proof. By Lemma 2, visible phase alone is not the full lifted state. A non-trivial lifted step therefore cannot be carried by only the two displayed Red registers; it also requires an independent sheet / guard carrier preserving the difference between visible witness and full carried state. That yields three independent burdens.

Any rooted binary tree with n internal nodes has $n + 1$ leaves. To carry three independent inputs, such a tree must therefore have at least three leaves and hence at least two internal nodes. A single internal EML node has only two leaves, so it cannot be state-complete. $\square$

Corollary 1 (Why the Red node is already composite). *The Red node can be a valid continuous quotient while still failing primitive status. Its two displayed arguments are not enough to carry the full burden of one non-trivial lifted step.*

6 Projector-preserved invariants

The quotient law does not merely preserve values. It preserves classes. Two classes matter here because they delimit the Red image sharply and cleanly.

Lemma 3 (Analyticity of the Red image). *Every finite EML tree is real-analytic on its natural domain D_T .*

Proof. The proof is by structural induction on the grammar $T ::= 1 \mid x_i \mid \text{eml}(T, T)$. The terminals 1 and x_i are real-analytic. If S and R are real-analytic on their natural domains, then $\exp(\llbracket S \rrbracket(x))$ is real-analytic and $\ln(\llbracket R \rrbracket(x))$ is real-analytic wherever $\llbracket R \rrbracket(x) > 0$. Their difference is therefore real-analytic on the parent natural domain. $\square$

Theorem 11 (Analyticity obstruction correspondence). *Real-analyticity on natural domains is a projector-preserved invariant of the Red image. Therefore $|x|$ on any open interval containing 0 lies outside $\text{Image}(\Pi_{\text{Red}})$.*

Proof. By Lemma 3, every finite EML tree is real-analytic on its natural domain. That analyticity class is therefore preserved by the Red image. The function $|x|$ is not differentiable at 0: its left derivative is -1 and its right derivative is $+1$. Therefore $|x|$ is not real-analytic on any neighborhood of 0, so it cannot lie in the Red image there. $\square$

Lemma 4 (Liouvillian closure of the Red image). *Every finite EML tree lies in the exp-log / Liouvillian closure of $\mathbb{C}(x)$.*

Proof. The terminals belong to $\mathbb{C}(x)$. The grammar adjoins only arithmetic combination, exponentiation, and logarithm. Finite closure under those operations is exactly the exp-log / Liouvillian construction of differential algebra (Ritt, 1948; Rosenlicht, 1972). $\square$

Theorem 12 (Liouvillian obstruction correspondence). *Liouvillian exp-log closure is a projector-preserved invariant of the Red image. Therefore roots of a generic irreducible quintic lie outside $\text{Image}(\Pi_{\text{Red}})$.*

Proof. By Lemma 4, every finite EML tree lies in the Liouvillian closure of $\mathbb{C}(x)$. That class is therefore preserved under the Red projector. Classical differential algebra places elementary exp-log constructions inside Liouvillian extensions (Ritt, 1948; Rosenlicht, 1972). Classical Galois theory gives that the generic irreducible quintic has Galois group S_5 , which is non-solvable. Its roots are therefore not Liouvillian. Since the Red image preserves the Liouvillian class, such roots lie outside the image. $\square$

7 Worked example: $\sin(x)$ descent

Theorem 13 (Lifted rotation to real readout). *On the declared principal branch, the lifted rotation witness descends to the real sine readout by the chain*

$$\widehat{\Xi}_{\text{rot}}(x) \xrightarrow{\text{Vis}} ie^{ix} \xrightarrow{\Pi_H} (\sin x, \cos x),$$

where

$$\widehat{\Xi}_{\text{rot}}(x) := \left(A_0, (1, 1), x, \left\lfloor \frac{x}{2\pi} \right\rfloor, [x - \pi, x + \pi] \right),$$

$$\Pi_H(ie^{ix}) = (-\text{Re}(ie^{ix}), \text{Im}(ie^{ix})) = (\sin x, \cos x),$$

and

$$e^{ix} = \text{Exp}_{\text{PC}}(ix) = \text{eml}_{\text{PC}}(ix, 1).$$

Consequently,

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i} = \frac{\text{eml}_{\text{PC}}(ix, 1) - \text{eml}_{\text{PC}}(-ix, 1)}{2i}$$

is a lawful descended witness formula.

Proof. The primitive roll gives the visible witness $\text{Vis}(\widehat{\Xi}_{\text{rot}}(x)) = ie^{ix}$ by direct substitution into $z(\theta) = ie^{i\theta}$. The real readout is immediate:

$$-\text{Re}(ie^{ix}) = -\text{Re}(i \cos x - \sin x) = \sin x, \quad \text{Im}(ie^{ix}) = \cos x.$$

The complex exponential is already present on the continuous-shadow branch, and by Theorem 7 the Red node is exactly $\text{eml}_{\text{PC}}(u, v) = e^u - \ln v$, so $e^{ix} = \text{eml}_{\text{PC}}(ix, 1)$ and $e^{-ix} = \text{eml}_{\text{PC}}(-ix, 1)$. Substituting the Euler identity yields the displayed witness for $\sin x$. $\square$

This worked example is the right kind of witness: not an appendix census, but a full descent from carried state to visible trigonometric readout.

8 Closure statement

Theorem 14 (Closure). *EML is a continuous-shadow composite and not a primitive constructor in the Phase Calculus dependency order. The primitive operator underneath it is the native selector evolution*

$$\text{PC}_{\text{op}}(\widehat{\Xi}) = \Sigma_{W, \Delta}(\widehat{\Xi})(\widehat{\Xi}).$$

Proof. Proposition 1 gives the direct execution witnesses for the native selector evolution. Theorem 6 places the operation layer below EML. Theorem 7 proves that the Red node is an exact quotient of lifted evolution. Theorem 10 shows that a single EML node is never state-complete for one non-trivial lifted step. Theorems 11 and 12 show that the Red image preserves decisive invariant classes rather than opening an unrestricted primitive surface. Theorem 13 gives an end-to-end descent of a nontrivial function. Therefore EML is not primitive. It is the continuous quotient shadow of a deeper carried-state grammar whose native selector evolution is already operational. $\square$

The final placement is therefore exact:

$$\begin{aligned}
 & \boxed{\text{PC}_{\text{op}}(\widehat{\Xi}) = \Sigma_{W,\Delta}(\widehat{\Xi})(\widehat{\Xi})} \\
 & \implies \boxed{\Pi_{\text{Red}} \circ E_{\text{lift}} = G_{\text{Red}} \circ \Pi_{\text{Red}}} \\
 & \implies \boxed{\text{eml}_{\text{PC}}(x, y) = \text{Sub}_{\text{PC}}(\text{Exp}_{\text{PC}}(x), \text{Log}_{\text{PC}}(y)) = e^x - \ln y}
 \end{aligned}$$

EML compresses a continuous calculator surface. Phase Calculus supplies the native lifted-state operator and the discrete engine underneath that compression.

A Key witnesses

The appendix is decisive rather than exhaustive. It records the smallest witness surface that shows the hierarchy without flattening the paper into a checklist.

Basis item	Witness formula in the descended stack	Source / role
1	terminal 1 / proto- $\mathbb{N}$ seed	Exists below continuous syntax.
e	$\text{Exp}_{\text{PC}}(1)$	Continuous-shadow exponential evaluated at the seed.
i	$z(0)$	Primitive source witness of the roll.
π	$\inf\{\theta > 0 : R_\theta = -I\}$	Native half-turn witness before EML syntax.
$\exp(x)$	$\text{Exp}_{\text{PC}}(x)$	Direct shadow exponential branch.
$\ln(x)$	$\text{Log}_{\text{PC}}(x), x > 0$	Positive real inverse branch of the exponential.
$\sqrt{x}$	$\text{Sqrt}_{\text{PC}}(x) = e^{\frac{1}{2} \ln x}, x \geq 0$	Root witness inherited from exp-log closure.
$x + y$	$\text{Add}_{\text{PC}}(x, y)$	Descended additive structure from arithmetic completion.
x^y	$\text{Pow}_{\text{PC}}(x, y) = e^{y \ln x}, x > 0$	Power witness on the positive real branch.
$\sin(x)$	$(e^{ix} - e^{-ix})/(2i)$	Worked descent from lifted rotation to real readout.

Table 2: Key witnesses only. The full 36-item calculator surface follows from [Theorem 9](#); the appendix records the decisive rather than exhaustive witness surface.

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